CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (ME) Examination

November-December 2012

CL 202 Strength of Materials

Date: 28.11.2012, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Derive: $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$, with usual notations. **[04]**

(b) It is required to design a close coiled helical spring which shall deflect 10 mm under an axial load of 200 N at a shear stress of 100 N/mm^2. The spring is to be made out of round wire having modulus of rigidity of 9×10^4 N/mm^2. The mean diameter of the coils is to be 10 times the diameter of the wire. Find the diameter and length of the wire necessary to form the spring. **[06]**

Q - 2 (a) A solid shaft of 200 mm diameter has the same cross-sectional area as that of a hollow shaft of the same material with inside diameter of 150 mm and the maximum shear stress developed in each of them are same. Find the ratio of the power transmitted by the shafts at the same speed. **[05]**

(b) A simply supported beam 3 m long has a section symmetrical about the y-y axis as shown in **Figure 1**. The beam carries two loads of equal magnitude as shown in **Figure 1**. If the maximum compressive and tensile stresses are not to exceed 100 N/mm^2 and 120 N/mm^2, determine the maximum value of W. **[06]**

OR

Q - 2 (a) A hollow shaft is to transmit 300 kW at 80 rpm. If the shear stress is not to exceed 60 N/mm^2 and the internal diameter is 0.6 of the external diameter, find the external and internal diameters, assuming the maximum torque is 1.4 times the average torque. **[05]**

(b) A flitched beam consists of a wooden joist 150 mm wide and 300 mm deep strengthen by a steel plate 12 mm thick and 300 mm deep on either side of the joist. If the maximum stress in the wooden joist is 7 N/mm^2. Find the corresponding maximum stress attained in steel. Find also the moment of resistance of the section. Take $E_s = 20\ E_w$. **[06]**

Q - 3 Attempt any Two **[14]**

(a) A rectangular block of material as shown in **Figure 2** is subjected to a tensile stress of 20 N/mm^2 on one plane and a tensile stress of 10 N/mm^2 on a plane at right angle, together with shear stresses of 10 N/mm^2 on the same planes. Find

(i) Magnitude and direction of the principal planes

(ii) Magnitude and direction of maximum shear stress

(b) What is Mohr's circle?

Draw Mohr's circle for the element subjected to the stresses as shown in **Figure 3** and find normal stresses, tangential stresses and resultant stresses on a plane inclined at 55° with vertical. Also find maximum tangential stress.

(c) For an element shown in **Figure 4** find

(i) Normal stresses, Tangential stresses and Resultant stresses on a plane inclined at 30° with vertical.

(ii) Maximum tangential stress

(iii)Magnitude and direction of the principal planes

SECTION – II

Q - 4 (a) Draw qualitative shear stress distribution diagram for following sections: **(Any Three)** **[03]**

(i) Diamond section

(ii) Symmetrical I section

(iii)T section

(iv)L section

(b) State and prove the parallel axis theorem. **[04]**

Q - 5 (a) Find moment of inertia of the shaded area as shown in **Figure 5** about AB axis. **[07]**

(b) A steel beam section as shown in **Figure 6** is subjected to a shear force of 600 kN. Determine the shear stress distribution for the beam section. **[07]**

OR

Q - 5 (a) Find moment of inertia of the section as shown in **Figure 7** about centroidal axis XX. **[07]**

(b) The beam section shown in **Figure 8** is subjected to a shear force of 40 kN. Sketch the shear stress distribution for the section. **[07]**

Q - 6 Attempt any Two [14]

(a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 9.**

(b) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 10.**

(c) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 11.**

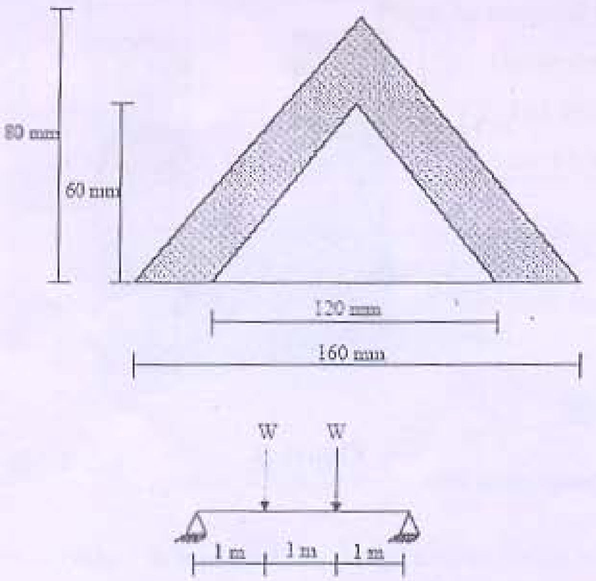

Figure 1

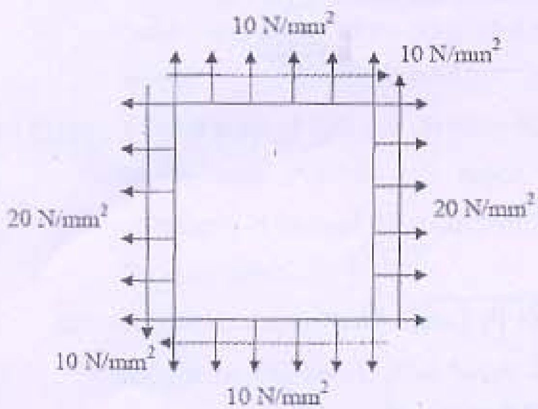

Figure 2

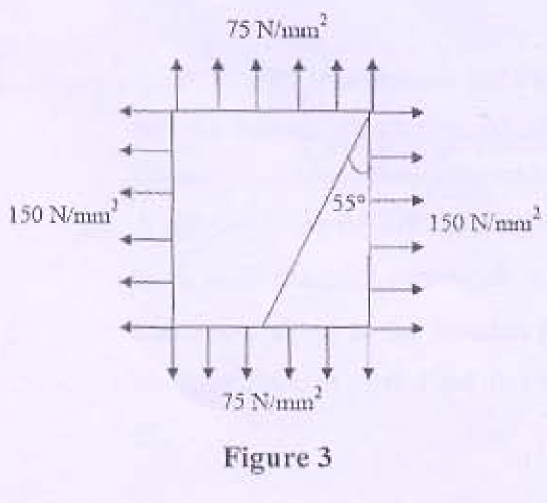

Figure 3

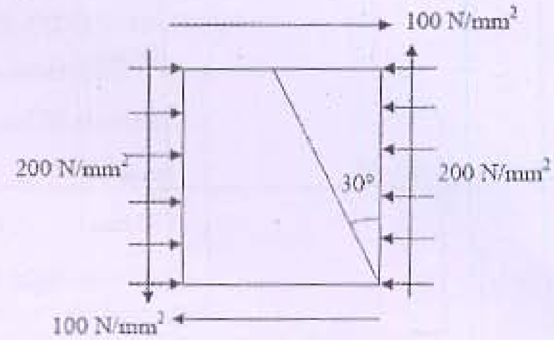

Figure 4

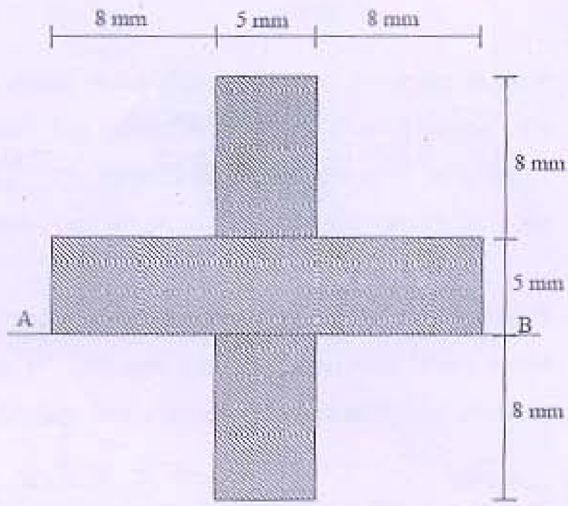

Figure 5

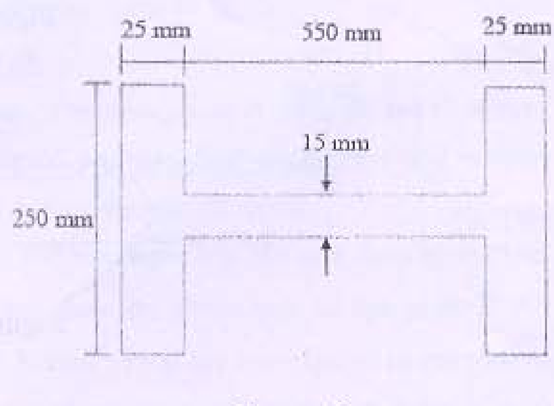

Figure 6

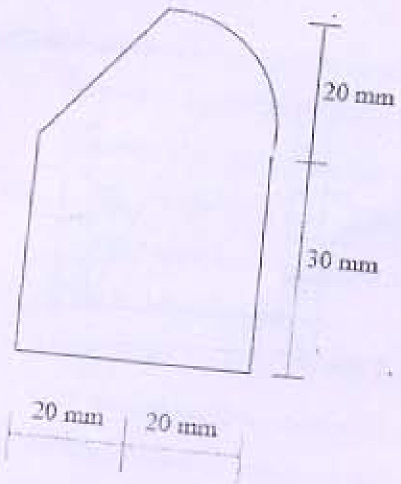

Figure 7

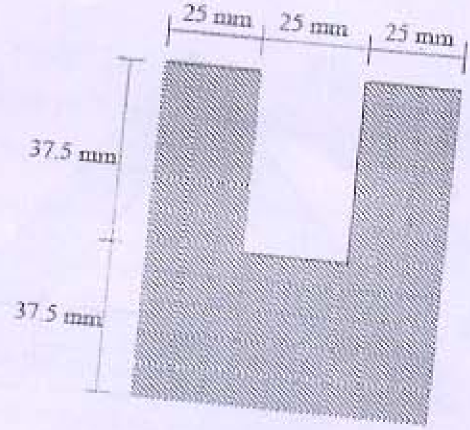

Figure 8

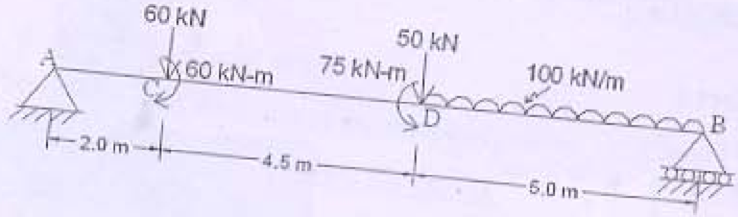

Figure 9

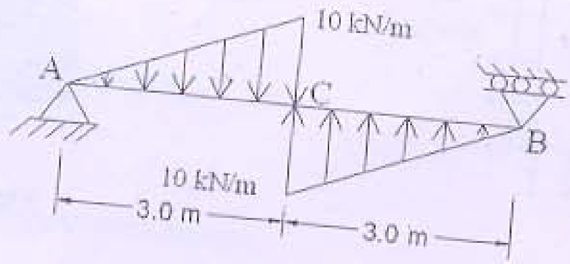

Figure 10

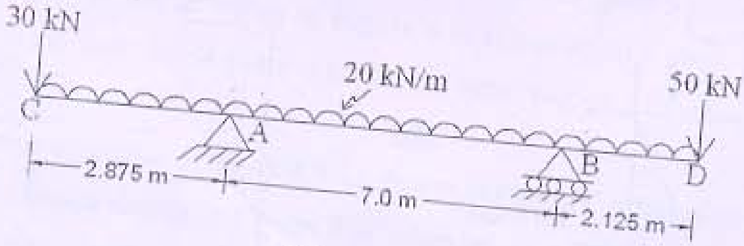

Figure 11

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